



Uptake of the cyanobacterial toxin microcystin by crop plants irrigated with contaminated wastewater: a review

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Abstract Microcystin, a potent cyanobacterial toxin, poses a potential health risk to humans through exposure to recycled water used for food crop irrigation. This comprehensive review synthesizes current knowledge on microcystin uptake and accumulation in various food crops, assessing associated health risks based on reported toxin levels and tolerable daily intake values. The impact of wastewater irrigation on microcystin uptake is also evaluated. Our analysis reveals critical knowledge gaps, highlighting key research priorities for ensuring food safety and mitigating human exposure to microcystin. Future studies should address these gaps to inform evidence-based guidelines for safe recycled water use in agriculture.

Keywords Microcystin · Cyanobacteria · Wastewater · Bioaccumulation · Food safety · Health risk

1 Introduction

Cyanobacteria are the oldest, most diverse group of microorganisms. They are Gram negative photosynthetic bacteria that play a vital role in the carbon and oxygen cycle, accounting for 20–30% of the Earth's photosynthetic productivity (Lee et al. 2017a; Breidenbach et al. 2022; Paerl and Paul 2012). However, cyanobacteria, commonly known as blue-green algae, also cause negative effects on human health and ecology by reducing water quality, producing odours and toxins and disrupting aquatic ecosystem structures and functions (Lee et al. 2017a; Lone et al. 2015; Maghsoudi et al. 2015; Cao et al. 2018). Microcystins (MCs), which are the most common toxin, can be produced by most bloom-forming cyanobacteria. Toxic effects of MCs on aquatic life have been well documented (Lee et al. 2017a; Lone et al. 2015), however, their effect on food crops is a relatively new area of research. The bioaccumulation of MCs, or any other toxin, in food crops, food animals, or aquatic products for human consumption is a potential human health risk due to the likelihood of these toxins being ingested (Lee et al. 2017a; Lone et al. 2015; Machado et al. 2017; Zhang et al. 2021; Liu et al. 2021). While there have been no known poisonings from consumption of MC contaminated food crops, many research papers have reported toxins in the edible parts of plants at levels that could result in exceedance of allowable daily intakes (Codd et al. 1999; Mohamed

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and Al Shehri 2009; Cao et al. 2018; Petrou et al. 2020.

Recycled water use is increasing worldwide and is largely used by the agricultural sector (Helmecke et al. 2020), however, cyanobacterial blooms within wastewater may affect its suitability for food crop irrigation. Prior to agricultural use, treatment facilities treat the wastewater to reduce biological or chemical contaminants. Wastewater treatment plants can be divided into 2 categories, conventional electrochemical wastewater treatment plants and wastewater stabilisation ponds (Romanis et al. 2021). Both types of wastewater treatment facilities remove biodegradable organic compounds, pathogenic organisms, and suspended solids from wastewater. Regardless of the type of wastewater treatment plant, the treatment of wastewater is critical to ensure that it is safe for crop irrigation. Unfortunately, wastewater storages are prone to intense proliferation of cyanobacteria due to highly nutrient rich waters (Romanis et al. 2021). However, the presence of cyanobacteria does not necessarily guarantee production of MCs, as not all species produce these secondary metabolites and production can be paused if the environment is not favourable (Romanis et al. 2021; Wood et al. 2011). Once MCs are present in a water source, they are difficult to remove through conventional wastewater treatment procedures. In a naturally occurring bloom, it is also likely that more than one toxin is present (e.g. a mixture of MCs), however, the concentration of certain toxins may predominate (Díez-Quijada et al. 2018).

The World Health Organisation (WHO) has determined a safety guideline for MCs in drinking water of $1.0 \mu\text{g L}^{-1}$ for long term exposure, with a provisional level of $12 \mu\text{g L}^{-1}$ for short term exposure (WHO 2017; Spoof et al. 2020; Abdallah et al. 2021; Campos et al. 2021). The WHO have also introduced a tolerable daily intake of MC-LR (microcystin leucine-arginine) which is $0.04 \mu\text{g kg}^{-1}$ of body weight. This total daily intake includes all MCs ingested from drinking water and food as well as recreational activities in contaminated water (Cao et al. 2018; WHO 2017; Campos et al. 2021). In waters used as irrigation sources, MCs have been found in concentrations between 4 and $50 \mu\text{g L}^{-1}$ up to $6500 \mu\text{g L}^{-1}$ (Corbel et al. 2014a). A recent study by Mohamed et al. (2024a) found MC concentrations in the Minia Governorate Egypt ranged from non-detectable to $93.7 \mu\text{g L}^{-1}$ and varied depending on the month of

sampling (Mohamed et al. 2024; Corbel et al. 2014a). Mohamed et al. (2024b) also showed that Sohag Governorate in Egypt (south of the sites used in the study above) showed a smaller range in MC concentration, between $1.4 \mu\text{g L}^{-1}$ and $15.6 \mu\text{g L}^{-1}$. However, showed that there was a decrease in winter and through the cooler months (Mohamed et al. 2024a). Redouane et al. (2023) found that their study water source, Lalla Takerkoust reservoir, had up to $1622 \mu\text{g mL}^{-1}$ of total MC content. Use of recycled water contaminated with toxins for irrigation may therefore present a harmful effect on the health of humans and livestock, but also crops and the soil ecosystem (Buratti et al. 2017; Machado et al. 2017; 2017; Mohamed et al. 2024b).

Recently, increasing human activity has led to eutrophication of freshwaters, in turn increasing occurrence and intensity of cyanobacterial blooms, approximately half of which are known to produce harmful metabolites (Machado et al. 2017; Pantelic et al. 2023; Buratti et al. 2017). Climate change has been projected to contribute to an increase in eutrophication and therefore excess cyanobacteria, which can degrade the ecosystems that are vital to humans Manubolu et al. 2018.

Current available literature on the effects of MCs on the growth and functionality of crop plants is based around plants being exposed to different concentrations of toxin with differing exposure lengths, with few common features between studies and often details lacking from methods, making it difficult to quantify the rate of toxin uptake from contaminated irrigation water into the plant (Table 1). Studies into wastewater as an irrigation source has also focused mainly on pathogens or heavy metals, and there is not a large focus on organic contaminants (Yan et al. 2021). This literature review aims to provide a look at the potential risks associated with using wastewater contaminated with MCs as a food crop irrigation source.

Literature for this review was sourced using Flinders University's online library, which provides access to multiple academic databases and publisher websites (such as Elsevier, Springer, Wiley), and Google Scholar. Articles were selected based on relevance to the topic of wastewater irrigation and its effects on soil properties, using search terms such as "wastewater irrigation," "cyanobacteria toxins," "salinity," and "microcystins". Approximately

Table 1 Showing total MC-LR or MC exposure over the course of experiments with the amount of toxin recovered reported from previous literature

Reference	Plant	MC Recovered ($\mu\text{g kg}^{-1}$)	Tissue	Growth Method	Total toxin exposure (μg)
Crush et al. (2008)	Lettuce	0.00079	Shoots	Pots	170
	Clover	0.0002	Shoots	Pots	170
Jarvenpaa et al. (2007)	Broccoli	2.2	Roots	Pots	190
Saqrane et al. (2009)	Pea	191,000 DW	Roots	Pots	420
	Pea	79,000 DW	Stem	Pots	420
	Pea	157,000 DW	Leaves	Pots	420
	Durum wheat	16,000 DW	Roots	Pots	420
	Durum wheat	1000 DW	Stem	Pots	420
	Durum wheat	16,000 DW	Leaves	Pots	420
Hereman and Bittencourt-Oliviera (2012)	Corn	not detected	Total tissue	Pots	420
	Lettuce	177.8	Foliar tissue	Pots	187.5
	Lettuce	16.8	Foliar tissue	Pots	93.45
Gutierrez-Praena et al. (2014)	Tomatoes	10.52–10.83	Fruits	Hydroponics	400
	Tomatoes	12,298.18	Young leaves (second and third pairs from apex)	Hydroponics	400
Corbel et al. (2015)	Tomatoes	21,600,000	Whole plant	Pots	9000
Corbel et al. (2016)	Tomatoes	not detected	Fruits	Pots	9000
	Tomatoes	304	Stems	Pots	9000
	Tomatoes	0.1035	Leaves	Pots	9000
	Tomatoes	0.3118	Roots	Pots	9000
	Tomatoes	0.3118	Roots	Pots	9000
Bittencourt-Oliveira et al. (2016)	Lettuce	158.35	Total Accumulation	Pots	195
Liang et al. (2016)	Rice	281	Roots	Hydroponics	3000
	Rice	241	Leaves	Hydroponics	3000
	Rice	32.5	Stems	Hydroponics	3000
	Rice	18.6	Roots	Hydroponics	7
	Rice	18.2	Leaves	Hydroponics	7
	Rice	13	Stems	Hydroponics	7
Drobac et al. (2017)	Capsicum	118	Fruits	Pots	6860
Cao et al. (2017)	Rice	37,840	Leaves	Hydroponics	5000
	Rice	41,620	Stem	Hydroponics	5000
	Rice	113,680	Roots	Hydroponics	5000
Lee et al. (2017b)	Lettuce	> 100	Roots	Pots	120
	Lettuce	> 60	Edible shoots	Pots	120
	Carrot	≥ 200	Edible root	Pots	120
	Carrot	> 50	Non-edible Shoots	Pots	120

Table 1 (continued)

Reference	Plant	MC Recovered ($\mu\text{g kg}^{-1}$)	Tissue	Growth Method	Total toxin exposure (μg)
Cao et al. (2018)	Rice	37.8	Root	Hydroponics	5000
	Rice	41.6	Stem	Hydroponics	5000
	Rice	114	Leaves	Hydroponics	5000
	Rice	80	Root	Hydroponics	500
	Rice	25	Stem	Hydroponics	500
	Rice	20	Leaves	Hydroponics	500
	Rice	22	Root	Hydroponics	50
	Rice	10	Stem	Hydroponics	50
	Rice	5	Leaves	Hydroponics	50
Wijewickrama and Manage (2019)	Rice (BG358)	567.52	Total tissue	Hydroponics	383,684.40
	Rice (Suwandel)	429.83	Total tissue	Hydroponics	383,684.40
	Water spinach	350.82	Total tissue	Hydroponics	383,684.40
Xiang et al. (2020)	Sweet potato	0.63–10.7	Total tissue	Pots	1500
	Brown mustard	0.24	Total tissue	Pots	1500
	Chinese broccoli	0.1–15	Total tissue	Pots	1500
Mohamed et al. (2022)	Common bean	8.6	Roots	Pots	528
	Common bean	12.2	Shoots	Pots	528
	Common bean	78.2	Immature pods	Pots	528
Cao et al. (2023)	Lettuce	102.7	Leaves	Pots	210
	Lettuce	111.8	Leaves	Pots	210
	Lettuce	58.1	Leaves	Pots	210
	Carrot	193.3	Root	Pots	280
	Carrot	219.7	Root	Pots	280
	Carrot	148	Root	Pots	280

Total toxin has been calculated using the toxin exposure per watering, and frequency of watering per experiment. Some experiments have been excluded from table due to insufficient methodological details

400 articles between 2010 and 2024 were initially reviewed, with older articles being reviewed where necessary.

2 The importance of wastewater

In water scarce times, it becomes necessary to consider alternative options for crop irrigation (Kesari et al 2021; Khalid et al. 2018; Jaramillo and Restrepo 2017). There has been increasing research into the potential of recycled wastewater as an important crop irrigation component. Khalid et al. (2018) believe that 60% of the world's total population will face water shortage by 2025, with Yan et al. (2021) believing

that around 40% of the population will face this issue (Khalid et al 2018; Yan et al. 2021). This is due to many factors such as pollution, population growth, weather conditions, climate change and rapid urbanisation (Yan et al. 2021, Musa Yahaya et al. 2023; Mishra et al. 2023). Wastewater provides many benefits such as the presence of organic matter and nutrients, which can enhance crop production; reduced contamination of natural bodies of water, which have historically been used as disposal points, as well as reduced reliance, and improved conservation, of primary water sources such as rivers and reservoirs (Jaramillo and Restrepo 2021; Khalid et al. 2018).

Wastewater has been used as an irrigation source since 3200 BC, when domestic wastewater would

be used in agriculture as a way of disposing this from urban settlements (Jaramillo and Restrepo 2017). This spread to farms across Germany, England and Scotland between 1550 and 1700. During the 1800 many fast-growing cities across the United States and Europe began to irrigate soils with wastewater and was considered as a disposal solution for large volumes of wastewater (Jaramillo and Restrepo 2017; 2018). In Australia, the first crop field to be irrigated with wastewater was in 1897 in Melbourne. Interest in using wastewater for agricultural purposed peaked in the 1990's due to the large water demand for agricultural lands, which is approximately 70–90% of available water sources (Musa Yahaya et al. 2023; Jaramillo and Restrepo 2017; Kesari et al. 2021; Mishra et al. 2023).

2.1 Wastewater reuse

In Australia, the 'Millennium Drought' which occurred between 2001 and 2009 forced the country to save water for drinking and use recycled water for non-drinking purposes (Radcliffe and Page 2020). The Werribee wastewater treatment plant (WWTP), which was built in 1897, as well as sites in South Australia including Bolivar and Glenelg WWTPs have been updated to a modern infrastructure capable of producing recycled water for industrial and agricultural use. New Haven Village in South Australia uses recycled water treated from wastewater for house gardens, toilet flushing and sports oval irrigation (Thomas et al. 2000; Radcliffe 2022; Radcliffe and Page 2020; Fearnley et al. 2004). State capitals are a major source of recycled water given the large population and large volumes of treated wastewater (Radcliffe 2022). Both Adelaide and Melbourne use recycled water for agricultural irrigation in areas close to their WWTPs (Radcliffe 2022).

In a review undertaken by Radcliffe and Page (2004), it was found that by 2001/2002, over 500 of Australia's wastewater treatment plants were recycling some or all treated wastewaters. Some of these sites were recycling the water internally, however, other uses for the recycled wastewater were irrigation of livestock pastures, sporting ovals, nursing home gardens, as well as irrigation of vegetables and vineyards (Radcliffe 2022).

3 Likelihood of cyanobacterial accumulation in wastewater

Cyanobacteria are ubiquitous in freshwater sources and occur in marine and brackish waters (Díez-Quijada et al. 2019; Fontanillo & Köhn, 2018; Gutiérrez-Praena et al. 2013; Mohamed et al. 2021; Omid et al. 2018; Zhou et al. 2015). They proliferate in rivers, lakes, or other water sources where eutrophication has occurred due to human activity and blooms usually occur between spring and autumn. Growth of cyanobacteria can be enhanced by high pH, low turbulence and water column stratification, high nitrogen and phosphorus levels, temperature > 18.8 °C as well as a light intensity > 26 $\mu\text{mol m}^{-2} \text{s}^{-1}$ (Garda et al. 2016; Jiang et al. 2008; Kumar et al. 2018; Zhou et al. 2015). The environment in wastewater treatment plants is ideal conditions for cyanobacteria blooms to grow, and in some management strategies, growing on non-toxic cyanobacteria species is done to help with bioremediation and for biofuel production (Romanis et al. 2021). Other used for non-toxic cyanobacteria include nutrient removal. Although the growth of non-toxic cyanobacteria is encouraged, this leads to the potential production of MCs with growth of MC producing cyanobacteria species such as *Microcystis* (Romanis et al. 2021). Although the presence of cyanobacteria blooms in wastewater treatment facilities is not indicative of the presence of cyanotoxins such as MCs, there is the potential for toxins to be produced by the cyanobacteria that are present and therefore pose a risk to human health as well as potential impacts on plants if used for irrigation (Romanis et al. 2021). To help mitigate this risk, there are usually treatment processes put in place to combat the cyanobacteria blooms as well as monitor the amount of cyanobacteria and or toxin concentrations (Romanis et al. 2021).

3.1 Cyanobacterial toxins

Cyanobacteria blooms can grow to large densities within days or weeks, and release secondary metabolites, some of which are called cyanotoxins due to their toxic effect on other organisms. MCs, which are cyclic heptapeptides and are classified as carcinogenic, are the most widespread of these cyanotoxins and can be found in freshwater, marine water and brackish water sources during cyanobacterial blooms

Pappas et al. 2020; Mohamed et al. 2024a; Garda et al. 2016; Kumar et al. 2018; Zhou et al. 2015 (Lee et al. 2017a; Lee et al. 2017b; Lone et al. 2015; Corbel et al. 2014a; Alhaithloul et al. 2024; Gácsi et al. 2009). MCs can be produced by more than 30 cyanobacteria species belonging to genus' such as *Microcystis*, *Dolichospermum* (*Anabaena*), *Planktothrix*, *Nostoc*, *Anabaenopsis*, *Aphanizomenon*, *Limnothrix*, *Phormidium*, *Oscillatoria*, *Rivularia* and *Hapalosiphon* (Abdullah et al. 2021; Babica et al. 2006; Fontanillo and Kohn 2018; Gutiérrez-Praena et al. 2014; Juhász et al. 2023; Kulabhusan and Campbell 2024; Rastogi et al. 2014; Zhu et al. 2018). It is thought that most MCs are intracellular while few (less than 30%) are extracellular (Buratti et al. 2017; Wu et al. 2015). In any bloom, the concentrations of intracellular MCs can vary up to 40-fold such as 100–4000 fg/cell in *Microcystis aeruginosa*, with higher numbers generally seen at the beginning of the bloom (Buratti et al. 2017). MCs are usually retained within healthy cyanobacteria cells during exponential growth phase, thus the concentration of dissolved MC within water is generally less than 10 µg L⁻¹. Intracellular MCs contained within the cell are released upon cell lysis (Buratti et al. 2017; Redouane et al. 2023).

There is a global problem of increasing MC content in bodies of water, partly due to increased water temperatures resulting from climate change, as well as anthropogenic eutrophication of water sources, both of which favour the blooms of toxin-forming cyanobacteria species (Cao et al. 2019; Mohamed et al. 2024a; Redouane et al. 2023). High levels (up to mg L⁻¹) of extracellular MCs can be found when algaecides are added to treat a bloom. The decay and collapse of a bloom causes the toxin to be released (Wan et al. 2019; Romanis et al. 2021). The half-life of the most common analogue, MC-LR ranges from several days to months depending on the environment. The half-life of MC-LR in soils has been reported to be between 6 and 17.1 days (Kumar et al. 2018; Miller and Fallowfield 2001; Redouane et al. 2019).

Reducing eutrophication of surface waters is the first step in controlling cyanobacteria proliferation and the toxins they produce “Pantelic et al. 2013”. These toxins have significant effects, not only on the ecosystem, but on human and animal health (Weralupitiya et al. 2022; Lee et al. 2017a; Mutoti et al. 2022; Pflugmacher et al. 2007). These toxins can spread

through different processes; accumulation in plants, absorption in soil, or leaching into groundwater in the environment. Toxins can then find their way into humans by ingestion of contaminated food (plant or animal) or water, skin contact, ingestion or inhalation during showering, bathing and recreation in contaminated water sources (Buratti et al. 2017; Corbel, Mougin et al. 2014; Machado et al. 2017; Weralupitiya et al. 2022).

3.2 Degradation of cyanobacterial toxins

MCs have presented many challenges to water treatment facilities as conventional treatment methods have limited effects on their removal (Pantelic et al. 2013). The best short-term control for *Microcystis* blooms is through the use of chemical algaecides which in turn removes the source of MCs. However, the subsequent release of high levels of MC can challenge conventional treatment plants and have negative impacts on aquatic ecosystems (Pantelic et al. 2013). Ishii et al. 2004 . Traditional water treatment methods can remove a small amount of extracellular MCs, however, the majority will be removed as intracellular, present in whole cells, via the coagulation process (Redouane et al. 2019). Chemical and biological inactivation include conventional technologies such as coagulation, sand filtration, chlorination and activated carbon absorption (some of which are currently used in Australian drinking water treatments) as well as advanced oxidation technologies such as ozonation, UV photolysis, Fenton oxidation and sonolysis (Pantelic et al. 2013). Oxidants, chlorine, ozone, hydroxyl radical, chloramine, potassium permanganate and chlorine dioxide, are commonly used as disinfectants before chemical addition or after filtration. Factors such as temperature, pH and treatment chemicals can change the demand for oxidant and lower the effectiveness of toxin inactivation Pantelic et al. (2013). Corbel et al. (2014a) reviewed papers from Jones and Orr (1994); Chen et al. (2008); Edwards et al. (2008); Harada et al. (1996); Sivonen and Jones (1999) that MCs can remain in aquatic ecosystems for weeks after being released and may be able to remain for several months or years under natural conditions (Chen et al. 2008; Corbel, Mougin et al. 2014; Edwards et al. 2008; Harada et al. 1996; Jones and Orr 1994; Sivonen and Jones 1999). Zastepa et al. (2014) backed up these claims with their findings

showing that MCs remained in the water column for up to 40 weeks after the collapse of a cyanobacterial bloom (Preece et al. 2017; Zastepa et al. 2014).

UV degradation may speed up their removal from water (Corbel et al. 2014b) and an adequate exposure of UV can also be used as a treatment itself, however, this can also be affected by wavelength, turbidity, UV dose, presence of organic matter and presence of chemical catalysts (titanium dioxide, hydrogen peroxide etc.). When UV is used with titanium dioxide and hydrogen peroxide as catalysts, the catalysts produce free radicals, which disrupt the structure of MCs (Lee et al. 2017a).

4 Irrigation of crops with contaminated wastewater

In areas where freshwater is in short supply, treated recycled wastewater could be used for food crop irrigation to maintain the drinking water supply for critical needs. Treated wastewater effluent is currently used to irrigate approximately 359,000 km² of croplands worldwide (mainly in developing countries), whereas raw or treated wastewater are used to irrigate over 20 million hectares of cultivated land (Jaramillo and Restrepo 2017; Kesari et al. 2021; Khalid et al. 2018; Yan et al. 2021). However, some areas have been using wastewater for crop irrigation despite not being legally permitted, and in other areas wastewater is considered an attractive source of irrigation given the lower cost (Khalid et al. 2018b; Mohammed Rusan et al. 2007; Yan et al. 2021). Use of raw wastewater is a high-risk form of irrigation. However, its widespread use can be attributed to the cost, preservation and nutrient value that is perceived in the areas where it is practiced. (Jaramillo and Restrepo 2017; Kesari et al. 2021; Liang et al. 2014; Mohammad Rusan et al. 2007; Yan et al. 2021). However, although the wastewater can provide nutrients, there is the potential for the exposure of harmful chemicals and pathogens, as well as the exposure to too many nutrients and harmful chemicals and pathogens (Kesari et al. 2021; Liang et al. 2014; Mohammad Rusan et al. 2007). Wastewater irrigation has been found to increase both soil salinity and heavy metal levels, while decreasing soil pH (Kesari et al. 2021; Yerli et al. 2024).

The World Health Organization drafted the “Reuse of effluents: methods of wastewater treatment and health safeguards” in 1973 with the aim of protecting public health and the facilitating the use of wastewater in agriculture (Jaramillo and Restrepo 2017). This document has been updated many times as more information surrounding the risks is being researched and understood, becoming the WHO’s 2006 guidelines for the safe use of wastewater, excreta and greywater. Other government agencies worldwide have also written and distributed guidelines for wastewater reuse (Jaramillo and Restrepo 2017).

Given the risks associated with untreated wastewater irrigation, treated wastewater has been recommended for use for use in agricultural regions, under controlled conditions, to minimise contaminants and pathogens in the soils and crops Liang et al 2014 (2007). One of the major concerns with irrigating food crop from recycled wastewater is the potential for it to transport toxins such as MCs. Existing regulations around wastewater reuse are directed at human health from microbial pathogens and heavy metals and do not typically include organic contaminants (Yan et al. 2021).

It is estimated that crops irrigated with wastewater have been consumed by approximately 10% of the world’s population (Jaramillo and Restrepo 2017; Jeong et al 2016; Kesari et al. 2021; Mishra et al 2023; Yan et al 2021). Liang et al. 2014 reported that a large amount of the wastewater used for irrigation is used indirectly by being distributed into water sources used for the irrigation of agricultural lands (Jaramillo and Restrepo 2017; Kesari et al. 2021; Liang et al 2014). Jaramillo and Restrepo (2017) also reported that approximately 400 m³ of raw wastewater is discharged into irrigation water sources in Latin America and then used to irrigate different crops (Jaramillo and Restrepo 2017).

Irrigation is essential for agricultural production, especially fruit and vegetable crops. The two common methods of irrigation are drip and spray irrigation, both with their benefits and potential risks or difficulties (Cao et al. 2018; Codd et al. 1999; Saqrane et al. 2009). Of these, spray irrigation is the most common method. Spray irrigation can utilise groundwater, however, surface water is most commonly used, and therefore susceptible to contamination during cyanobacteria blooms (rbel et al. 2015; Corbel et al 2014; ElKhalloufi et al. 2013; Ferrao-Filho

et al 2014; Pflugmacher et al. 2007). The use of contaminated water for spray irrigation of crops leads to the potential for these crops to be contaminated with MCs (Bittencourt-Oliveira et al. 2016; Drobac et al. 2017), in concentrations that could exceed the tolerable daily intake (TDI) limit of $0.04 \mu\text{g kg}^{-1}$ body weight (Bittencourt-Oliveira et al. 2016; WHO 2017).

A study by Tsoumalakou et al. (2021) compared the use of spray and drip irrigation systems to determine which system would result in the least MC contamination in spinach plants (Tsoumalakou et al. 2021). Spinach (*Spinacia oleracea* L. var. *Virofly*) plants were grown in 2L pots using a soil and perlite mixture and were exposed to either tap water or water from Karla's Reservoir in Greece containing $3.8 \mu\text{g L}^{-1}$ of MCs. Separate sets of plants were watered by drip or spray irrigation twice a week, to determine the differences in spinach quality and MC concentration. This study found that the spinach plants exposed to MCs via either irrigation method had delayed germination and final growth, which were significantly lower than the tap water controls. They also reported that spray irrigating with water containing MCs decreased the product quality and safety for human consumption due to toxin concentration, while drip irrigation of water containing MCs decreased growth, stomatal density and size as well as interfering with photosynthetic characteristics (Tsoumalakou et al. 2021). It has also been reported that using water contaminated with *M. aeruginosa* to spray irrigate lettuce plants results in the cyanobacteria cells becoming lodged in the leaves up to 10 days after the irrigation Corbel et al. 2014a.

Previous studies have presented the possibility of removing MCs from plants through purging (Machado et al. 2017). Cordeiro-Araújo et al. (2016) found that lettuce plants exposed to 50 and $10 \mu\text{g L}^{-1}$ MC-LR for 7 days could be irrigated for a further 29–37 days with non-contaminated water to eliminate the toxin from within the lettuce plant. Although this strategy became less effective at higher concentrations of toxin (Cordeiro-Araújo et al. 2016; Spoof et al. 2020). In addition, due to the growth period of lettuce, 50–70 days, purging would not be possible as a management practice unless the toxin was detected within the first 30 days of planting lettuce. Arugula (*Eruca vesicaria*; edible rocket) plants were not investigated for purging as the accumulation of MC-LR

in the Arugula plants was below the detection limit Cordeiro-Araújo et al. 2016 (2021).

5 MC-LR in soil systems

Due to the difficulty in extracting total MCs from soil, there has been limited research in determining MC content in soils with the majority of work investigating processes impacting MC levels in water (Pham and Utsumi 2018). Minimal research has also been conducted into water treatment residual (WTR) in bloom affected areas, and we are unaware of research on the impact of WTR on the soil microbiome. The soil microbiome helps to maintain soil health as well as crop development and performance (Ai et al. 2020; Mutoti et al. 2022; Lee et al. 2017a; Liu et al. 2021).

Toxins may be retained in soil and bioaccumulate in the soil during non-bloom seasons (Ai et al. 2020; ee et al. 2017a; Liu et al. 2021; Mutoti et al. 2022 (L). Toxins can come into contact with the soil via irrigation of the plants with contaminated water, degradation of cyanobacterial cells within the soil, flooding by a contaminated water source or by application of contaminated fertiliser (Ding et al. 2020; Xiang et al. 2020). In countries such as China, agriculture fields border contaminated water sources, causing MC contamination of these soils at concentrations up to $381.4 \mu\text{g kg}^{-1}$ (Cao et al. 2022). A higher toxin concentration in soil is generally as a result of irrigation with water containing a high level of toxin (Cao et al. 2017). MCs have also been found in dried cyanobacterial cells, suggesting they can re-enter the soil when re-immersed in water (Corbel et al. 2014b; Corbel et al. 2014a). Once MCs enter the soil, they can remain there with a half-life of several days to several months, dependant on degradation efficiency (Kumar et al. 2018; Redouane et al. 2019; Xiang et al. 2014; Cao et al. 2018; 2018; Cao et al. 2018b).

Another area lacking research is the adsorption of MCs in soil. So far, it has been suggested that adsorption of MCs is more significant in clay soils and non-existent in sandy soils (Miller and Fallowfield 2001; Miller et al. 2001). It has been shown that soils with high organic carbon content (2.9%) and high clay content (16.1%) were most effective at removing toxins, whereas it was difficult to remove toxins from sandy soils (Corbel et al. 2014b; Machado et al. 2017). It has been suggested that the reason for better

adsorption of MCs in clay soils is due to the ability to bind to the metal ions in clay. This adsorption capacity can also be decreased with acidic pH (5.6) when MCs and clay particles display a negative charge (Cao et al. 2017; Chen et al. 2006; Corbel et al. 2016).

Researchers have shown that MCs can inhibit soil microbial activity and reduce nitrification processes with potential negative impacts on plant health (Cao et al. 2017; Redouane et al. 2019; Liu et al. 2021). MCs can also be dissipated in soils; however, this is controlled by microbial and physiochemical properties of the soil. Soil microbiota experiments measuring soil respiration, enzyme activity (acid phosphatase, alkaline phosphatase, β -glucosidase, dehydrogenase, urease phenol oxidase) reported that dehydrogenase activity (relevant to microbial respiration) increased, and phenol oxidase activity was slightly inhibited by MCs (Redouane et al. 2019). A study by Corbel et al. (2015b) investigated the enzymatic activities and potential nitrification by looking at alterations in sulphur, phosphorus, and nitrogen mineralisation and cellulose degradation. The activity of four soil enzymes, arylsulfatase, phosphatase, urease, and β -D-glucosidase were shown not to have been significantly affected after 14 days of irrigation with MC-LR contaminated water (Corbel et al. 2015b).

In a study by Cao et al. (2018b), lettuce and rice (*Oryza sativa*) plants treated with low ($2.61 \mu\text{g L}^{-1}$ on average) and high ($5.22 \mu\text{g L}^{-1}$ on average) concentrations of MCs showed toxin accumulation in the soil, leaves and roots to concentrations similar to those measured in the irrigation water; however, it was not detected in the soil after harvest (Cao et al. 2018). The lack of MCs present in the soil after harvest was attributed to degradation within a 30-day period. Cao et al. (2018) mentioned a previous, unpublished study using the same soil, which showed MCs eliminated within 18 days (Cao et al. 2018). There was also no significant difference in the soil concentrations between the lower concentration and higher concentration treatments, which was also shown by Levizou et al. (2017) who reported soils accumulated approximately 1.7–4.7% of what plants accumulated, a similar result to those in Corbel et al. (2016) (Cao et al. 2018a; Corbel et al. 2016; Levizou et al. 2017). These studies contradict the results of Lee et al. (2017b), which found that MCs can persist in soil for up to 8 weeks after crop plants have been harvested, even

with no additional watering of contaminated water. Soil used in Corbel et al. (2016) contained 76% sand, 13% silt and 11% clay and soil used in Cao et al. (2018b) contained 53% sand, 31.3% silt, and 15.5% clay were relatively similar in composition. Whereas soil used in Levizou et al. (2017) was listed as peat with no further information, and soil used in Lee et al. (2017b) was purchased and comprises of perlite, bark, Canadian peat moss, dolomite lime, among other ingredients, however an exact breakdown is not available, making it difficult to compare studies (Lee et al. 2017; Levizou et al. 2017; Cao et al. 2018).

Once toxins have entered the soil they can be removed by degradation via UV, adsorption or biodegradation (Corbel et al. 2014b; Miller and Fallowfield 2001). However, the most common degradation method of toxin in soils is microbial Corbel et al. 2014b. Each type or group of toxins react quite differently when entering a soil ecosystem. Xiang et al. (2020) reported that the soils of plants irrigated with MC-LR contaminated water had concentration increases at first, however, concentrations decreased during the growth phase of plants. After 10 days of growth only approximately 10% of the total MC content added remained in soils, supporting the root uptake pathway hypothesis (Xiang et al. 2020). It was proposed that the continual input of MCs into the soil accompanied by microbial degradation of MCs could account for the characteristic concentration profiles (Xiang et al. 2020).

Toxins accumulating in a soil ecosystem may be difficult to overcome. These toxins can persist, depending on the efficiency of degradation in a particular soil. MCs can take longer to degrade due to their half-life range of several days to months (reported to be from 6 days up to 90 days) (Corbel et al. 2014b; Kumar et al. 2018; Redouane et al. 2019; Xiang et al. 2014). However, the data available about MCs in soil systems is scarce and conflicting; more studies are needed to be able to accurately determine the soil systems role on food crop uptake of MCs (Corbel et al. 2014a).

6 Plant exposure pathways, uptake and transport

Many diverse types of vegetables, grains, legumes and even some fruits have been studied to determine uptake, transport, and accumulation of MCs within

plants. From these studies, it has been determined that the length and frequency of toxin exposure affects the degree of bioaccumulation (Lee et al. 2017a).

6.1 Exposure pathways

Irrigation methods mentioned previously give MCs an exposure pathway into the plant as well as soil, however, there can be issues with discerning between translocation from the roots to the shoots, compared to uptake through the stomata in the leaves (Bartos 2020, Corbel et al. 2014a). Spray irrigation is common for agriculture and horticulture, which may cause surface contamination of the soils and plant (Bouaicha 2016; Pflugmacher et al. 2007). However, the usual exposure pathway mentioned in literature is MCs being absorbed from soil into roots then translocated to the shoots (Corbel et al. 2014a; Machado et al. 2017). Corbel et al. (2014a) noted that the uptake pathway in vegetable plants is relatively unknown, unlike the pathway within vertebrates, however, it is thought that they may cross cell membranes by diffusion or root absorption and then translocated to the shoots (Corbel et al. 2014a; Chen et al. 2012). Specific transporters have not yet been described, however, several different membrane transporters with affinities for amino acids and different peptides have been considered (Machado et al. 2017).

A study by Pflugmacher et al. (2019) reported that in semi-hydroponically grown wheat seedlings spray irrigated with water spiked with MC-LR the uptake was time dependent and occurred via the root pathway (Pflugmacher et al. 2019). At a dose of $0.1 \mu\text{g L}^{-1}$, approximately $1.4 \mu\text{g g}^{-1}$ of MC-LR was found in the roots 2 h after exposure to the toxin, which exponentially increased to a maximum concentration of approximately $69.6 \mu\text{g g}^{-1}$ after 168 h of exposure. Pflugmacher et al. (2019) assumed that due to the polar, water-soluble nature of MCs, they had entered the root by being dissolved in water via the apoplastic pathway.

The rhizospheric microbiota development has also been influenced by the presence of MCs (Cao et al. 2017; El Khalloufi et al. 2016; Reedouane et al. 2019). It has been found that rhizosphere microbiota plays a role in helping to avoid the adverse impacts of MCs on plants. A study by Redouane et al. (2021) used *T. aestivum* L., var. *Achta* (common wheat) and *V. faba* L., var. *Alfia* 321 (fava bean) and grew

these seeds in sterile and non-sterile soils and irrigated with water containing either $0 \mu\text{g L}^{-1}$ or $100 \mu\text{g L}^{-1}$ of MC-LR every second day as well as irrigating both soils without plants for 30 days (Redouane et al. 2021). Toxin concentrations were determined by enzyme-linked immunosorbent assay (ELISA) in both plant tissue and soils. The study found that the root system for both common wheat and fava beans were highly sensitive to MCs in sterile soils compared to the control group, decreasing lateral root number and fasciculated root number. It was also found that at $100 \mu\text{g L}^{-1}$ of MC-LR exposure in both sterile and non-sterile soils, there were no detrimental effects on the growth of common wheat and fava beans, indicating that the natural soil microbiota can protect these plants from MC-LR (Redouane et al. 2021).

6.2 Root vs stomatal uptake

A 2021 study by Lee et al. reported that cyanobacteria could be internalised by any opening site whether natural such as stomata or damaged areas as an insect bite or human handling Lee et al. (2021). Lee et al. (2021) applied 1 mL of *M. aeruginosa* solution to lettuce leaves 3 times per week between weeks 4 and 8 of growth. They then tested for accumulation, productivity and human health risk of the lettuce plants. Lee et al. (2021) managed to identify *M. aeruginosa* cells between the guard cell using SEM, thus providing evidence of potential stomatal uptake (Lee et al. 2021).

Pflugmacher et al. (2019), exposed wheat seedlings to the toxin via the roots showing that uptake and translocation does occur within plants Pflugmacher et al. (2019). This is supported by the findings of Xiang et al. (2020), who showed that MC-LR concentrations in the roots were 1–2 times that of the concentrations in the stems and leaves. The study used both drip and spray irrigation and found no evidence of stomatal uptake of MC-LR and determined that MCs were absorbed mainly by the roots and then translocated by the transpiration stream (Xiang et al. 2020).

6.3 Transport

Currently, the root-to-shoot transport hypothesis is supported by the detection of MCs in leaves, with data also suggesting that the accumulation within the

plant is dependent on the concentration and time of exposure (Azevedo et al. 2013; Lee et al. 2017a; Cao et al. 2018; Machado et al. 2017).

It has previously been reported that the main mode of transport within the reed plant *Phragmites australis*, was through the stem and rhizome, and uptake through the leaves may occur during contact with contaminated water (Corbel et al. 2014a). Pham and Utsumi (2018) noted that MCs are unable to penetrate vegetable cell membranes, and that MCs accumulate directly via the root adsorption of dissolved MCs, diffusion or potentially by enzymatic formation of MC conjugates, via glutathione-S-transferase (Pham and Utsumi 2018). Gutiérrez-Praena et al. (2014) proposed two hypotheses regarding MC uptake and toxin translocation from roots to leaves and fruits after seeing a decrease in MCs detected in the fruits during the second week of MC exposure. The first hypothesis was the ability for the toxin to be chemically metabolised or modified in plant tissues. There is growing evidence to support this hypothesis: MCs can form conjugates with glutathione either spontaneously or via activity of glutathione-S-transferase, however, the toxin may also become inaccessible due to potentially binding to other biomolecules (Gutiérrez-Praena et al. 2014). Another hypothesis is that the toxin is deposited in a vacuole or in the apoplast, a mechanism which protects cells from toxic compounds (Gutiérrez-Praena et al. 2014).

Alternatively, Pflugmacher et al. (2019) proposed a symplastic route for MC transport through the plant, due to the hydrophobicity of the Casparian strip, forcing dissolved MCs across the plasma membrane before entering the xylem (Pflugmacher et al. 2019).

7 Accumulation

Bioaccumulation is the process where MC concentrations become higher inside the organism or tissue compared to the surrounding environment (Cord-eiro-Araujo et al. 2016; Gutiérrez-Praena et al. 2013). There have been many studies on the accumulation and uptake of MCs in plants. Levizou et al. (2020) noted that most recent studies into the bioaccumulation of MCs show that the toxin was generally found in higher concentrations within the roots than the leaves. However, these studies focused on vegetables where the leaves were the edible parts of the

vegetable. These findings are consistent with reported toxin levels found in Table 1 of this review.

Although there are only a small number of studies investigating the MC accumulation in crop plants and soils, the studies have shown clearly that MCs can accumulate in edible plant tissues after irrigation with contaminated water (Lee et al. 2017), as shown in Table 1. However, in some of these studies there was limited information on irrigation method and MC concentration in the irrigation water, uptake into crop plants and soils, uptake differences between tissues and persistence in soils after harvesting (Lee et al. 2017).

A study by Cao et al. (2018b) treated rice and lettuce plants with a low concentration of MCs (MC-LR, MC-RR and MC-YR) ($2.61 \mu\text{g L}^{-1}$) and a high concentration of MCs ($5.22 \mu\text{g L}^{-1}$) to determine their effects on the plant and soils. This study reported that the lettuce plants exposed to the high concentration of MC-LR had a leaf concentration of 3–4 times higher than those exposed to the low concentration. Interestingly, there was a higher concentration in the lettuce roots than were measured in the lettuce leaves of both treatments (Cao et al. 2018b). Rice showed a similar pattern to lettuce, and there was also a small MC-LR accumulation in the grains ($1.4\text{--}5.2 \mu\text{g kg}^{-1}$ or 0.2% of total MCs found in rice plants) compared to the rest of the plant (highest measurement being $1504 \mu\text{g kg}^{-1}$ in roots). Rice also had higher concentrations than the lettuce plants (up to $424 \mu\text{g kg}^{-1}$ in roots), likely because the rice plants had longer exposure to the contaminated water. This study found that the accumulation by the high concentration treatment resulted in the accumulation order of: roots > leaves > grains $\approx$ soils, while the low concentration resulted in: roots $\approx$ leaves > grains $\approx$ soils (Cao et al. 2018b).

Peuthert et al. (2017) studied green pea (*Pisum sativum* (L.) Koern), sugar pea (*Pisum sativum* vr. *saccharatum* (L.) Koern.), chickpea (*Cicer arietinum* (L.)), green and red mung bean (*Vigna radiata* (L.) R. Wilcz.), French bean (*Phaseolus vulgaris* (L.)), soya bean (*Glycine max* (L.) Merrill.), alfalfa (*Medicago sativa* L.), lentil (*Lens culinaris* Medik.), maize (*Zea mays* L.), and wheat (*Triticum aestivum* L.) and found that the highest MC-LR concentration was measured in the roots of alfalfa and wheat. This study only exposed the roots to the toxins and determined that the toxin was transported into the shoot and primary

leaves, which is common in research into uptake and transport within plants (Peuthert et al. 2017).

Levizou et al. (2020) investigated the bioaccumulation in edible roots of carrot (*Daucus carota*) and radish (*Raphanus sativus*). This study used naturally contaminated water from Karla's Reservoir in Greece, which had a constant concentration of $3.76 \mu\text{g L}^{-1}$ of MCs ($2.03 \mu\text{g L}^{-1}$ MC-LR and $1.43 \mu\text{g L}^{-1}$ MC-RR) for the duration of the research. Radish was grown in soil for 1.5 months with 4 treatment groups differing in exposure start times. Carrots were grown for 5 months with 3 treatment groups. It was found that radish plants, which received the contaminated water from seed were approximately 50% smaller than the control radishes, the taproots were also smaller in volume compared to the control. Carrots that were exposed to the contaminated water source from seed were approximately 70% smaller than the control as well as a decrease in taproot volume (Levizou et al. 2020).

In studies with tomato plants, MCs were detected in the fruits, roots, and leaves, with lowest concentrations being detected in unripe tomatoes (Gutierrez-Praena et al. 2014). The pattern of accumulation in the tomato plants tested was leaves ($27,673.21 \mu\text{g kg}^{-1}$) > roots ($1635.21 \mu\text{g kg}^{-1}$) > mature tomato fruits ($10.52\text{--}10.83 \mu\text{g kg}^{-1}$) when exposed to a concentration of $100 \mu\text{g L}^{-1}$ every 3rd day for approximately 14 days (Gutierrez-Praena et al. 2014). However, it was also found that during the second week of exposure, concentrations decreased within the tomato fruits. Given the current state of knowledge, it is not easily explained (Gutierrez-Praena et al. 2014). A study by Romero-Oliva et al. (2014) found that tomato and capsicum plants bioaccumulate MC-RR in their fruits and seeds when irrigated with contaminated water (Abdullah et al. 2021; Romero-Oliva et al. 2014, 2021; 2020).

Many studies have shown that the accumulation of MCs in food crop plants is dependent on several factors including dose of toxin, species of plant, exposure route, exposure length, and the type of tissues (Pindihama and Gitari 2020; Hereman and Bittencourt-Oliveira 2012). Recently, meta-analysis has shown that bioaccumulation of MCs in leafy vegetables such as cabbage, dill, and parsley were nearly 3 times higher than concentrations found in other plants (Abdallah et al. 2021; Campos et al. 2021).

8 Adverse effects on plants

Effects that MCs have on plants are dependent on purity of toxin or extract used, plant species, stage of development during which the exposure occurs, time of exposure and concentrations of toxin or extract (Freitas et al. 2015; Redouane et al. 2019; Machado et al. 2017). There is currently conflicting evidence on whether plant growth and yield are affected by MC uptake. Some studies have found no reported effects on plant growth, yield or germination (Campos et al. 2021; Janiele et al. 2021; Lee et al. 2017a; Cao et al. 2018), while others find a decrease. Some researchers have noted that studies reporting decreases in plant growth or yield used short exposure times (30 days or less) and purified toxins (Corbel et al. 2014b; Cao et al. 2018).

MCs potentially inhibit germination, growth and nodulation of plants (Azevedo et al. 2013; Zhu et al. 2018; Machado et al. 2017; Juhász et al. 2023). Nodulation inhibition may lead to an impact of nitrogen uptake and the development of legume plants. Differences in the root development and growth pattern emerge when exposed to $0.5\text{--}40 \mu\text{g L}^{-1}$ of MCs and may occur due to effects on the microtubule organisation (Azevedo et al. 2013). It has also been determined that root mitosis was inhibited at 2 days of exposure to $10\text{--}40 \mu\text{g L}^{-1}$ of MCs and completely inhibited after 20 days exposure to $20\text{--}40 \mu\text{g L}^{-1}$ (Azevedo et al. 2013).

MCs inhibit protein phosphatases 1 and 2A, which are critical for numerous biochemical and physiological processes such as mitosis, root growth and development as well as regulating key enzymes in carbon and nitrogen metabolism. This can cause activation of enzymes involved with CO_2 fixation, starch storage, gene expression, carbon and nitrogen metabolism, tissue development, alteration of chlorophyll content, photosynthesis as well as causing mitochondrial damage (Amorim et al. 2017; Chen et al. 2010; Corbel et al. 2014a; Janiele et al. 2021; Li et al. 2001; Machado et al. 2017; Pappas et al. 2020; Pflugmacher et al. 2007; Wan et al. 2019; Freitas et al. 2015b; Juhász 2023).

Bioaccumulation of MC can occur in edible tissues, leaves, roots, fruits, grains, and stems. It has been reported that a high accumulation of MC within tissues can potentially lead to a decrease in plant quality, however, there is conflicting evidence

(Cordeiro-Araujo et al 2016; Drobac 2017; Gutierrez-Praena 2013). Crush et al. (2008) found that shoot dry weight (DW) in ryegrass, rape and lettuce were reduced after 3 applications of lake water directly onto roots. After 6 applications of lake water containing $152,200 \text{ cells mL}^{-1}$ of *M. aeruginosa*, only lettuce showed reduction in shoot DW. After 3 applications of lake water directly onto roots, root DW of ryegrass (*Lolium perenne*) and rape (*Brassica napa*) were heavier than control. After 6 applications of lake water to roots, ryegrass DW was less than control ryegrass, but rape was still heavier. Clover (*Trifolium repens*) and lettuce did not differ. Application of lake water to shoots had no effects on shoot DW after 3 applications but after 6, ryegrass and lettuce shoots weighed less than the controls (Crush et al 2008).

Many studies have investigated photosynthetic and biochemical effects of plants after exposure to MCs. MCs have been shown to induce reactive oxygen species (ROS) accumulation and alter antioxidant systems (Freitas et al 2015, Mathe et al 2009; Xiang et al. 2020; Cao et al. 2018). These studies have reported changes in ROS, which are indicators of oxidative stress and activate antioxidants such as superoxide dismutase (SOD) and peroxidase (POD) activities (Chen et al. 2010; El Khalloufi et al 2013). Azevedo et al. (2013) used MC-LR extracted from *M. aeruginosa* and found no change in growth, photosynthetic efficiency and glutathione S-transferase (GST) activities of rice seedlings exposed to MCs for up to 12 days, however, there was an increase in the antioxidant activity of glutathione peroxidase (GPx) in roots during a 2-day exposure to both $0.26 \mu\text{g L}^{-1}$ and $13 \mu\text{g L}^{-1}$ (Azevedo et al. 2013, 2021). After 2 days exposure, Prieto et al. (2011) reported that rice plants had similar leaf levels of GST and GPx to the control plants, but after longer exposure (6–9 days), GST and GPx increased in roots and GST also increased in leaves during exposure to $50 \mu\text{g L}^{-1}$ of MC-LR (Prieto et al 2011).

Gutiérrez-Praena et al. (2014) reported during their study on tomato plants that there was a decrease in MC-LR levels during the second week of growth. Two and a half month old tomato plants were exposed to $100 \mu\text{g L}^{-1}$ MC-LR (water was spiked with either $100 \mu\text{g L}^{-1}$ of pure toxin or $100 \mu\text{g L}^{-1}$ of extracted toxin) every 3 days for two weeks. During the second week of exposure, it was noted that the MC content within tomatoes fell below detection limit of

$0.58 \mu\text{g L}^{-1}$, this was seen in all samples except for the water that was spiked with extracted MC (Gutiérrez-Praena et al 2014).

9 Human exposure

Humans may be exposed to MCs via ingestion, contact with contaminated recreation water or inhalation during recreational activities (Azevedo et al. 2013; Buratti et al. 2017; Lee et al. 2017b; Mutoti et al. 2022; Pinho et al. 2015; Weralupitiya et al. 2022; Zhang et al. 2021; Miller 2001). Humans can ingest cyanobacterial toxins either by drinking contaminated water or potentially, by the ingestion of contaminated food including vegetables, fruit, and meat (Abdallah et al. 2021; Fontanillo & Köhn, 2018; Juhász et al. 2023; Campos et al. 2021). Ingestion of a high concentration of MCs can occur due to eating large quantities of contaminated foods such as seafood, animal products and crops, even when the initial concentration of toxin in the environment is low (Drobac et al 2013; Mutoti et al 2022; Larson et al. 2014).

MCs and other hepatotoxins are not able to penetrate the cell membranes of vertebrates and therefore must be taken up via other routes. Generally, the MCs are transported through the bile acid transport system and small intestines (Corbel et al. 2014a). MCs can be absorbed through the small intestines, travel via the hepatic portal system and then accumulate in the liver. From the liver, with the help of organic anionic transporting polypeptides (OATP) for bile salts, MCs cross the hepatocyte cell membranes. When inside the hepatocyte cells, MCs covalently bind to protein phosphatases 1 and 2a, in turn disrupting hepatic architecture (Fontanillo & Köhn, 2018; Gutiérrez-Praena et al. 2013; Juhász et al. 2023; Pappas et al. 2020; Žegura et al. 2011; Biales et al. 2020).

MCs can induce apoptosis in a variety of cells including lymphocytes, hepatocytes, neurons and kidney epithelial cells. Through the cleavage of intermediate filament proteins, apoptosis alters the integrity of the cytoskeleton (Huang et al 2015).

Cyanobacterial toxins are known to cause detrimental short-term and long-term health effects to humans, including tingling, numbness, burning, dermatitis, incoherent speech, gastrointestinal pain,

haemorrhage, inflammation, pneumonia, liver failure leading to death, skin tumours, cardiac arrhythmia leading to death, tumour promotion, potentially cancers, neurodegenerative disease, as well as irreversible damage to Ser/Thr protein phosphatase 1 and 2A (Buratti et al. 2017; Lee et al. 2017b; Mutoti et al. 2022; Sangolkar et al. 2006; Xiang et al. 2014; Lee et al. 2017a; Liu et al. 2021, 2014a; Mohamed et al. 2024b). On a cellular level, MC has been shown to stimulate ROS formation and cause DNA damage, and long-term exposure can cause loss of stability and function of mitochondrial DNA (mDNA) (Lee et al. 2017a).

The LD₅₀ (lethal dose that would kill 50% of population) is 50–56 µg kg⁻¹ of body weight (Jarvenpaa 2007; Xie et al. 2021; Bouaïcha et al. 2019). LD₅₀ measurements have largely been studied in mice models with early work using unpurified cyanobacterial extracts. Elleman et al. (1978) showed that intraperitoneal injection of mice with *M. aeruginosa* led to a LD₁₀₀ between 15–30 mg lyophilised algal bloom kg⁻¹ body weight. Purified toxin from the same extract was estimated to have an LD₅₀ of 56 µg kg⁻¹ and an LD₁₀₀ of 70 µg kg⁻¹ with parenteral administration, with death of mice occurring within 1–3 h (Bouaïcha et al. 2019; Elleman et al. 1978).

The WHO provisional TDI guideline value for chronic exposure is derived from a study by Fawell et al. 1999 where a 40 µg/kg bw per day for 13 weeks was administered by gavage to mice, with no treatment related effects. Incorporation of uncertainty factors results in a TDI for chronic exposure of 0.04 µg/kg bw per day (WHO 2020).

Abdullahi et al. (2022) studied spinach (*Spinacia oleracea*), cabbage (*Brassica oleracea*), lettuce, kenaf (*Hibiscus cannabinus*), bitter leaf (*Vernonia amygdalina*), carrot and tomato (*Lycopersicon esculentum*) plants that were grown in farming land and received water from a stream contaminated with MC. MC values for spinach, lettuce, tomato, bitter leaf and carrot were all greater than the recommended 0.04 µg of toxin per kg of body weight. It was also found that leafy vegetables such as spinach and lettuce generally would be more of a risk to humans due to bioaccumulation of MCs in the edible leaves (Abdullahi et al. 2022).

As yet, there have been no reported cases of poisoning from ingestion of contaminated crop plants, although other poisoning events have been reported

Bouaïcha 2016. These include two events in Brazil, namely MC intoxication of 126 dialysis patients of which 50 patients died (Pouria et al. 1998; Redouane et al. 2023), and a water contamination issue, which found gastrointestinal issues in 2000 people, 800 of whom died. Incidents of liver and colorectal cancers and liver damage have also been reported in Australia and China, both related to MCs (Azevedo et al. 2013; Corbel, Bouaïcha et al. 2015; Fontanillo & Köhn, 2018; Meneely & Elliott, 2013; Mutoti et al. 2022; Pantelic et al. 2013; Buratti et al. 2017; Bolo-taolo et al. 2020). Most poisoning incidences occur from prolonged ingestion of contaminated water at low concentrations (Drobac et al. 2013; Peuthert et al. 2007). While there have been many reported cases of disease such as cancers and organ damage from MCs, epidemiological studies can be difficult due to confounding factors, such as social or travel dynamics as well as climate or other pathogens, and the cause cannot always be determined (Farmer 2007; Communications 2018).

10 Health risk from food crop consumption

Review of the literature has identified studies which provide evidence of MC uptake by edible parts of irrigated crops with some researchers noting that the levels measured exceed the TDI for microcystins. However, a systematic approach is needed to properly evaluate what levels constitute a human health risk. Furthermore, this will also allow evaluation of the health risk based on existing literature data.

One way of assessing the risk of exposure to toxins via food crops is to determine the Maximum Allowable Microcystin-LR Content (MAMC) that can be present in a food crop before the Tolerable Daily Intake (TDI) is exceeded. To determine MAMC the following is needed:

- (1) Consumption patterns for the particular crop.
- (2) Tolerable daily intakes (TDI) for chronic exposure to MC.

The Australian Bureau of Statistics conducted an Australian Health Survey in 2011/2012, which provides a snapshot of food consumption patterns in Australia (Statistics 2014). This survey provides median daily consumption patterns for specific food groups

which can be used to determine a MAMC. Due to large variability in consumption patterns of fruit and vegetables in the community, the median, rather than mean, is a better measure of the central tendency of a group. The TDI for microcystin-LR is 0.04 µg/kg body weight/day which equates to 2800 ng/day for a 70 kg adult. Hence, the Maximum Allowable Microcystin-LR Content (MAMC) in various common food crops before exceedance of TDI based on Australian consumption patterns was calculated as follows and is presented in Table 2.

$$\begin{array}{|c|c|c|} \hline \text{Median Daily Consumption of} \\ \text{chosen food crop for an Adult (g)} \\ \text{(ABS, 2014)} & \div & \text{TDI for 70kg adult if sole} \\ & & \text{source of Microcystin =} \\ & & \text{2800ng/day (WHO, 2020)} \\ \hline & = & \text{Maximum Allowable Microcystin} \\ & & \text{Concentration (MAMC) in} \\ & & \text{food crop before exceedance of TDI for a} \\ & & \text{70kg adult (ng/g)} \\ \hline \end{array}$$

Toxin levels measured in various food crop studies were reviewed and compared to these MAMC values. Table 3 consolidates the data from these crop studies by separating into five groups in terms of growth medium and irrigation method:

1. **Field-** crop grown on a plot of land and irrigated with water containing MC.
2. **Laboratory hydroponics-** crops grown in water based nutrient solution containing MC.
3. **Laboratory, soil, irrigated at base-** crop grown in soil, contained in pots, irrigated with water containing MS at plant base.
4. **Laboratory, soil, irrigated over whole plant-** crop grown in soil, contained in pots, irrigated with water containing MC over whole plant (e.g., spray).
5. **Soil conditioner-** crop grown in soil supplemented with cyanobacteria biomass containing MC.

The review of the data and determination of MAMC exceedance is based solely on the toxin levels identified in the edible components of the food crops under the test conditions used by each researcher.

From Table 3, it can be seen that the vast majority of research publications investigating the bioaccumulation of MCs in food crops have been conducted in a laboratory or greenhouse setting. Only four publications investigated uptake of MCs into field grown crops including root vegetables, leafy greens, rice and fruiting vegetables. All four publications reported crops exceeding the MAMC, specifically leafy greens and rice. A total of three papers investigated toxin uptake by hydroponically grown crops. Hydroponics is seen as having the greatest potential for uptake of MC due to the direct contact of plant root with MC and lack of degradation processes that can occur in soil. However, only one study involving leafy greens recorded toxin levels high enough to exceed MAMC. There were ten studies where crops were watered at the base of soil grown plants propagated in pots. Two of the studies used soils that were already contaminated with MCs prior to planting as well as being irrigated with water containing MCs (Xiang et al. 2020; Ai et al. 2020) which would have added a further toxin loading to the system. Five of the studies used pure toxin in the irrigation water (Gutiérrez-Praena et al. 2014; Lee et al. 2017b; Peuthert et al. 2007; Xiang et al. 2020; Petrou et al. 2020), five with cyanobacteria extracts (Crush et al. 2008; Gutiérrez-Praena et al. 2014; S. Lee et al. 2017b; Peuthert et al. 2007; Petrou et al. 2020), and four used natural bloom material Ai et al. 2020; Levizou et al. 2017; Wijewickrama & Manage, 2019; Xiang et al. 2020) (Petrou et al. 2020; Pham and Utsumi 2018; Ai et al. 2020). Toxin concentration in food crops exceeded MAMC for four of the publications. (Petrou et al. (2020) in radish, Lee et al. (2017b) in carrots, Levizou et al. (2017) in lettuce, and Wijewickrama and Manage (2019) in

Table 2 Maximum allowable microcystin-LR content (MAMC) in particular food crops before TDI guideline value is exceeded

Food crop	Median daily consumption for an adult (g)	Maximum allowable microcystin concentration (MAMC) in food crop before exceedance of TDI for a 70 kg adult (ng/g)
Leaf and stalk vegetables	24.6	114
Root vegetables	37.2	75
Tomato	38	74
Rice	167.5	17

Table 3 Summary of publications investigating uptake and concentration of microcystin in edible components of food crops and exceedance of MAMC

Study type	No. of papers and reference details	Plant type tested	Number of papers reporting microcystin content exceeding of MAMC	Plant type showing exceedance of MAMC
Field	5 Codd et al. (1999), Mohamed and Al Shehri (2009), Xiang et al. (2019), Wijewickrama and Manage (2019) Mohamed et al. (2024a)	Root vegetables Leafy greens Rice Fruiting vegetables	5	Leafy greens Rice Potato
Laboratory Hydroponics	2 Cao et al. (2018), Llana-Ruiz-Cabello et al. 2019	Leafy greens	1	Leafy green
Laboratory, soil, irrigated at base	11 Peuthert et al. (2007), Järvenpää et al. (2007), Crush et al. (2008), Lefebvre (2013), Gutierrez-Praena et al. (2014), Lee et al. 2017b, Levizhou et al. (2017), Wijewickram and Manage 2019, Petrou et al. (2020), Xiang et al. (2020), Cao et al. (2023)	Leafy greens Fruiting vegetables Root vegetables Rice	5	Root vegetable Rice
Laboratory, soil, irrigated over whole plant	7 Crush et al. (2008), Hereman et al. 2012, Bittencourt-Oliveira et al. (2016), Cordeiro-Araújo et al. (2016), . (2016), Lee et al. (2017b), Lee et al. (2021), Cao et al. (2023)	Leafy greens Root vegetables	4	Leafy green Root vegetable
Soil conditioner	1 Xiang et al. (2020)	Leafy greens	0	

rice. A total of seven publications applied microcystin containing water by either pouring over the plant or as a spray. Lee et al. (2017b), Bittencourt-Oliveira et al. (2016), Hereman et al. (2012), and Crush et al. (2008) used a cyanobacterial culture extract of toxin in the irrigation water, and Cordeiro-Araújo et al. (2016), Lee et al. 2017b and Cordeiro-Araújo et al. (2016) used purified toxin. Four investigations identified toxin levels in the edible component in exceedance of MAMC. Lee et al. (2017b) for carrots, and Bittencourt-Oliveira et al. (2016), Hereman et al. (2012) and Crush et al. (2008) for lettuce. One study looking at MCs in a soil conditioner is presented here in which leafy greens were the test plant with levels not exceeding MAMC (Bittencourt-Oliveira et al. 2016; Cordeiro-Araújo et al. 2016; Crush et al. 2008;

Hereman & Bittencourt-Oliveira 2012; Lee et al. 2017b; Levizou et al. 2017; Wijewickrama & Manage, 2019; Petrou et al. 2020).

11 Conclusion and future perspectives

The uptake of microcystins (MCs) by crops entering the human food chain has not yet been comprehensively investigated, and most studies implement varying treatment strategies, making it difficult to draw firm conclusions. Many questions remain, particularly regarding whether this constitutes a potential risk to human health and whether crops irrigated with recycled water contaminated with MCs can be safely consumed. (Buratti et al. 2017; Corbel et al.

2014a; Machado et al. 2017). More research needs to be conducted on soil systems and the interaction between MCs, soils, and soil microbiomes to expand the current knowledge base. (Cao et al. 2018). Additional studies would provide a better understanding of which crops can be safely irrigated with recycled water, the impacts on crops themselves, and the potential risks to human health.

This review also emphasizes the importance of identifying approaches to manage the risks associated with human exposure to contaminated crops—whether through ingestion, handling, or cultivation. These management strategies could include optimizing irrigation methods, such as determining the relative safety of spray versus drip irrigation, and timing irrigation relative to the growth and development of edible plant tissues. Further understanding of soil type and microbiome influences on toxin adsorption, degradation, and transport to plant root zones would also enhance safety practices. Laboratory studies dominate the literature, necessitating more field studies to unravel the complex factors influencing MC uptake, translocation, and accumulation in crop plants for human consumption.

Furthermore, there is a need for focused research on the use of contaminated wastewater for irrigation. Current studies primarily address microbial pathogens and heavy metals, with limited attention to organic contaminants such as microcystins. It is critical to develop evidence-based guidelines for the safe use of recycled water in agriculture, addressing these organic toxins comprehensively. Several specific future research directions are identified: investigating interactions between MCs and soil microbiomes to develop effective bioremediation techniques; evaluating the efficiency of various irrigation methods, particularly spray versus drip irrigation, to minimize MC uptake by crops; assessing the effectiveness of post-harvest processing methods, such as washing and cooking, in reducing MC levels in crops; conducting field-based studies to mirror real-world agricultural practices and better assess risks under practical conditions; exploring climate change's role in exacerbating cyanobacterial blooms and its implications for wastewater irrigation.

Finally, understanding the broader community use of wastewater containing toxic cyanobacteria is essential for targeted implementation of guidelines. Addressing these knowledge gaps will ensure the safe

reuse of wastewater in agriculture while minimizing risks to human health and promoting sustainable agricultural practices.

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Data availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest S.F. is receiving an industry funded scholarship through Flinders University to complete her PhD. This literature review has been completed as part of her PhD.

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References

- Abdallah MF et al (2021) Cyanotoxins and food contamination in developing countries: review of their types, toxicity, analysis, occurrence and mitigation strategies. *Toxins* 13(11):786
- Abdullahi H et al (2022) Assessment of microcystins in surface water and irrigated vegetables in Kwaru stream, Hayin Danmani, Kaduna-Nigeria. *Environ Sci Pollut Res Int* 29:78303–78313
- Ai Y, Lee S, Lee J (2020) Drinking water treatment residuals from cyanobacteria bloom-affected areas: Investigation of potential impact on agricultural land application. *Sci Total Environ* 706:135756
- Alhaithloul HAS et al (2024) Corrigendum: Performance evaluation of *Moringa oleifera* seeds aqueous extract for removing *Microcystis aeruginosa* and microcystins from municipal treated-water. *Front Bioeng Biotechnol* 12:1487259
- Amorim CA, Ulisses C, Moura AN (2017) Biometric and physiological responses of *Egeria densa* planch.

- Cultivated with toxic and non-toxic strains of Microcystis. *Aquatic Toxicol* 191:201–208
- Azevedo CC et al (2013) Early physiological and biochemical responses of rice seedlings to low concentration of microcystin-LR. *Ecotoxicology* 23(2):107–121
- Babica P, Bláha L, Maršálek B (2006) Exploring the natural role of microcystins—a review of effects on photoautotrophic organisms. *J Phycol* 42(1):9–20
- Bartos AD (2020) Harmful algal blooms: dominance in lakes and risk for cyanotoxin exposure in food crops. ProQuest Dissertations Publishing
- Biales AD et al (2020) Global transcriptomic profiling of microcystin-LR or -RR treated hepatocytes (HepaRG). *Toxicol X* 8:100060–100060
- Bittencourt-Oliveira MC et al (2016) Lettuce irrigated with contaminated water: Photosynthetic effects, antioxidative response and bioaccumulation of microcystin congeners. *Ecotoxicol Environ Saf* 128:83–90
- Bolotaolo M et al (2020) Analysis of covalently bound microcystins in sediments and clam tissue in the Sacramento-San Joaquin River Delta, California, USA. *Toxins* 12(3):178
- Bouaïcha N (2016) Cyanobacterial toxins emerging contaminants in soils : a review of sources, fate and impacts on ecosystems, plants and animal and human health. *IntechOpen* 21:105–126
- Bouaïcha N et al (2019) Structural diversity, characterization and toxicology of microcystins. *Toxins* 11(12):714
- Breidenbach JD et al (2022) Microcystin-LR aerosol induces inflammatory responses in healthy human primary airway epithelium. *Environ Int* 169:107531–107531
- Buratti FM et al (2017) Cyanotoxins: producing organisms, occurrence, toxicity, mechanism of action and human health toxicological risk evaluation. *Archiv Toxicol Archiv Für Toxikologie* 91(3):1049–1130
- Cao Q et al (2023) Effect of different irrigation methods on the toxicity and bioavailability of microcystin-LR to lettuce and carrot. *Environ Sci Pollut Res Int* 30(47):104554–62. doi:10.1007/s11356-023-29800-2.
- Campos A et al (2021) Impacts of microcystins on morphological and physiological parameters of agricultural plants: A review. *Plants (Basel)* 10(4):639
- Cao Q et al (2018) Bioaccumulation of microcystin congeners in soil-plant system and human health risk assessment: a field study from Lake Taihu region of China. *Environ Pollut* (240):44–50
- Cao Q et al (2022) Using soil amendments to reduce microcystin-LR bioaccumulation in lettuce. *Environ Pollut* (292):118354–118354
- Cao Q et al (2017) Toxicological and biochemical responses of the earthworm *Eisenia fetida* to cyanobacteria toxins. *Sci Rep* 7(1):15954–15959
- Cao Q et al (2017) Effects of microcystins contamination on soil enzyme activities and microbial community in two typical lakeside soils. *Environ Pollut* 231:134–142
- Cao Q et al (2018a) Combined toxicity of microcystin-LR and copper on lettuce (*Lactuca sativa* L.). *Chemosphere* 206:474–482
- Cao Q et al (2018b) Effect of microcystins on root growth, oxidative response, and exudation of rice (*Oryza sativa*). *Ecotoxicol Environ Saf* 149:143–149
- Cao Q et al (2019) Bioaccumulation and detoxication of microcystin-LR in three submerged macrophytes: the important role of glutathione biosynthesis. *Chemosphere* 225:935–942
- Chen W et al (2006) Sorption, degradation and mobility of microcystins in Chinese agriculture soils: risk assessment for groundwater protection. *Environ Pollut* 144(3):752–758
- Chen W et al (2008) Reduction in microcystin concentrations in large and shallow lakes: water and sediment-interface contributions. *Water Res* 42(3):763–773
- Chen J et al (2010) Bioaccumulation of microcystin and its oxidative stress in the apple (*Malus pumila*). *Ecotoxicol* 19(4):796–803
- Chen J et al (2012) Accumulation and phytotoxicity of microcystin-LR in rice (*Oryza sativa*). *Ecotoxicol Environ Saf* 76(1):193–199
- Codd GA, Metcalf JS, Beattie KA (1999) Retention of Microcystis aeruginosa and microcystin by salad lettuce (*Lactuca sativa*) after spray irrigation with water containing cyanobacteria. *Toxicol* 37(8):1181–1185
- Communications N (2018) Epidemiology is a science of high importance. *Nature communications* 9(1) 1703–2
- Corbel S et al (2015a) Soil irrigation with water and toxic cyanobacterial microcystins accelerates tomato development. *Environ Chem Lett* 13(4):447–452
- Corbel S et al (2015b) Evaluation of phytotoxicity and ecotoxicity potentials of a cyanobacterial extract containing microcystins under realistic environmental concentrations and in a soil-plant system. *Chemosphere* 128:332–340
- Corbel S et al (2016) Evaluation of the transfer and the accumulation of microcystins in tomato (*Solanum lycopersicum* cultivar MicroTom) tissues using a cyanobacterial extract containing microcystins and the radiolabeled microcystin-LR (14C-MC-LR). *Sci Total Environ* 541:1052–1058
- Corbel S, Bouaïcha N, Mougin C (2014b) Dynamics of the toxic cyanobacterial microcystin-leucine-arginine peptide in agricultural soil. *Environ Chem Lett* 12(4):535–541
- Corbel S, Mougin C, Bouaïcha N (2014a) Cyanobacterial toxins: Modes of actions, fate in aquatic and soil ecosystems, phytotoxicity and bioaccumulation in agricultural crops. *Chemosphere* 96:1–15
- Cordeiro-Araújo MK et al (2016) Microcystin-LR bioaccumulation and depuration kinetics in lettuce and arugula: Human health risk assessment. *Sci Total Environ* 566–567:1379–1386
- Crush JR et al (2008) Effect of irrigation with lake water containing microcystins on microcystin content and growth of ryegrass, clover, rape, and lettuce. *Environ Toxicol* 23(4):246–252
- Ding Q et al (2020) Effects of microcystin-LR on metabolic functions and structure succession of sediment bacterial community under anaerobic conditions. *Toxins* 12(3):183
- Drobac D et al (2013) Human exposure to cyanotoxins and their effects on health. *Arh Hig Rada Toksikol* 64(2):119–130
- Drobac D et al (2017) Microcystin accumulation and potential effects on antioxidant capacity of leaves and fruits

- of *Capsicum annuum*. J Toxicol Environ Health-Part A-Curr Issues 80(3):145–154
- Díez-Quijada L et al (2018) New method for simultaneous determination of microcystins and cylindrospermopsin in vegetable matrices by SPE-UPLC-MS/MS. Toxins 10(10):406
- Díez-Quijada L et al (2019) Microcystin-RR: Occurrence, content in water and food and toxicological studies. Rev Environ Res 168:467–489
- Edwards C et al (2008) Biodegradation of microcystins and nodularin in freshwaters. Chemosphere 73(8):1315–1321
- Elleman TC et al (1978) Isolation, characterization and pathology of the toxin from a *Microcystis aeruginosa* (Anacystis cyanea) bloom [alga; stock water]. Aust J Biol Sci 31(3):209–218
- Farmer R (2007) The problems with some epidemiological studies. Maturitas 57(1):11–15
- Fearnley EJ et al (2004) Determination of water usage rates and water usage patterns in a residential recycling initiative in South Australia. Environ Health 4(2):72–81
- Ferrão-Filho AS, Herrera NA, Echeverri LF (2014) Microcystin accumulation in cladocerans: first evidence of MC uptake from aqueous extracts of a natural bloom sample. Toxicon 87:26–31
- Fontanillo M, Köhn M (2018) Microcystins: Synthesis and structure–activity relationship studies toward PP1 and PP2A. Bioorg Med Chem 26(6):1118–1126
- Freitas M et al (2015a) Lettuce (*Lactuca sativa* L.) leaf-proteome profiles after exposure to cylindrospermopsin and a microcystin-LR/cylindrospermopsin mixture: a concentration-dependent response. Phytochemistry 110:91–103
- Freitas M et al (2015b) Effects of microcystin-LR, cylindrospermopsin and a microcystin-LR/cylindrospermopsin mixture on growth, oxidative stress and mineral content in lettuce plants (*Lactuca sativa* L.). Ecotoxicol Environ Saf 116:59–67
- Garda T et al (2016) Microcystin-LR induces mitotic spindle assembly disorders in *Vicia faba* by protein phosphatase inhibition and not reactive oxygen species induction. J Plant Physiol 199:1–11
- Gutiérrez-Praena D et al (2013) Presence and bioaccumulation of microcystins and cylindrospermopsin in food and the effectiveness of some cooking techniques at decreasing their concentrations: a review. Food Chem Toxicol 53:139–152
- Gutiérrez-Praena D et al (2014) Exposure of *Lycopersicon Esculentum* to microcystin-LR: effects in the leaf proteome and toxin translocation from water to leaves and fruits. Toxins 6(6):1837–1854
- Gácsi M et al (2009) Comparative study of cyanotoxins affecting cytoskeletal and chromatin structures in CHO-K1 cells. Toxicol in Vitro 23(4):710–718
- Harada KI et al (1996) Stability of microcystins from cyanobacteria - III Effect of pH and temperature. Phycologia 35:83–88
- Harada KI et al (2020) Regulating water reuse for agricultural irrigation: risks related to organic micro-contaminants. Environmental sciences Europe 32(1):1–10
- Hereman TC, Bittencourt-Oliveira MC (2012) Bioaccumulation of Microcystins in Lettuce. J Phycol 48(6):1535–1537
- Huang X et al (2015) Involvement of oxidative stress and cytoskeletal disruption in microcystin-induced apoptosis in CIK cells. Aquat Toxicol 165:41–50
- Ishii H, Nishijima M, Abe T (2004) Characterization of degradation process of cyanobacterial hepatotoxins by a gram-negative aerobic bacterium. Water Res 38(11):2667–2676
- Janiele FN et al (2021) Effects of cyanotoxins in germination, growth and DNA damages of Lettuce sativa (Lettuce) and Coriandrum sativum (Coriander). Afr J Agric Res 17(1):39–45
- Jaramillo MF, Restrepo I (2017) Wastewater reuse in agriculture: a review about its limitations and benefits. Sustainability 9(10):1734
- Jeong H et al (2016) Assessing the effects of indirect wastewater reuse on paddy irrigation in the Osan River watershed in Korea using the SWAT model. Agric Water Manag 163:393–402
- Jiang Y et al (2008) Statistical study on the effects of environmental factors on the growth and microcystins production of bloom-forming cyanobacterium—*Microcystis aeruginosa*. Harmful Algae 7(2):127–136
- Jones GJ, Orr PT (1994) Release and degradation of microcystin following algicide treatment of a *Microcystis aeruginosa* bloom in a recreational lake, as determined by HPLC and protein phosphatase inhibition assay. Water Res 28(4):871–876
- Juhász GP et al (2023) Microcystin-LR, a cyanobacterial toxin, induces changes in the organization of membrane compartments in Arabidopsis. Microorganisms (Basel) 11(3):586
- Järvenpää S, Lundberg-Niinistö C, Spoof L, Sjövall O, Tyystjärvi E, Meriluoto J (2007) Effects of microcystins on broccoli and mustard, and analysis of accumulated toxin by liquid chromatography–mass spectrometry. Toxicon 49(6):865–874
- Kesari KK et al (2021) Wastewater treatment and reuse: a review of its applications and health implications. Water, Air, Soil Pollut 232(5):1–28
- Khalid S et al (2018) A review of environmental contamination and health risk assessment of wastewater use for crop irrigation with a focus on low and high-income countries. Int J Environ Res Public Health 15(5):895
- El Khalloufi F et al (2013) Physiological and antioxidant responses of *Medicago sativa*-rhizobia symbiosis to cyanobacterial toxins (Microcystins) exposure. Toxicon 76:167–177
- El Khalloufi F et al (2016) Microbiote shift in the *Medicago sativa* rhizosphere in response to cyanotoxins extract exposure. Sci Total Environ 539:135–142
- Kulabhusan PK, Campbell K (2024) Physico-chemical treatments for the removal of cyanotoxins from drinking water: current challenges and future trends. Sci Total Environ 917:170078–170078
- Kumar P et al (2018) Physico-chemical treatment for the degradation of cyanotoxins with emphasis on drinking water treatment—How far have we come? J Environ Chem Eng 6(4):5369–5388

- Larson D, Ahlgren G, Willén E (2014) Bioaccumulation of microcystins in the food web: a field study of four Swedish lakes. *Inland Waters* 4(1):91–104
- Lee S et al (2017b) Fresh produce and their soils accumulate cyanotoxins from irrigation water: implications for public health and food security. *Food Res Int* 102:234–245
- Lee S, Kim J, Lee J (2021) Colonization of toxic cyanobacteria on the surface and inside of leafy green: a hidden source of cyanotoxin production and exposure. *Food Microbiol* 94:103655
- Lee J, Lee S, Jiang X (2017a) Cyanobacterial toxins in freshwater and food: important sources of exposure to humans. *Annu Rev Food Sci Technol* 8(1):281–304
- Lefebvre BR et al (2013) The accumulation of the cyanobacterial toxin, microcystin, in cherry tomato (*Solanum lycopersicum*) and bush bean (*Phaseolus vulgaris*) plants. *Freshwater Biol Res* 15(1):1–11
- Levizou E et al (2017) Lettuce facing microcystins-rich irrigation water at different developmental stages: effects on plant performance and microcystins bioaccumulation. *Ecotoxicol Environ Saf* 143:193–200
- Levizou E et al (2020) Root vegetables bioaccumulate microcystins-LR in a developmental stage-dependent manner under realistic exposure scenario: the case of carrot and radish. *Agric Water Manag* 240:106274
- Li X, Liu Y, Song L (2001) Cytological alterations in isolated hepatocytes from common carp (*Cyprinus carpio* L.) exposed to microcystin-LR. *Environ Toxicol* 16(4):517–522
- Liang Q et al (2014) Long-term effects of irrigation using water from the river receiving treated industrial wastewater on soil organic carbon fractions and enzyme activities. *Agric Water Manag* 135:100–108
- Liang C et al (2016) The accumulation of the cyanobacterial toxin, microcystin, in cherry tomato (*Solanum lycopersicum*) and bush bean (*Phaseolus vulgaris*) plants. *Freshwater Biol Res* 15(1):1–11
- Liu B-L et al (2021) Variant-specific adsorption, desorption, and dissipation of microcystin toxins in surface soil. *J Agric Food Chem* 69(40):11825–11834
- Lone Y, Koiri RK, Bhide M (2015) An overview of the toxic effect of potential human carcinogen Microcystin-LR on testis. *Toxicol Rep* 2:289–296
- Machado J et al (2017) Effects of microcystin-LR and cylindrospermopsin on plant-soil systems: a review of their relevance for agricultural plant quality and public health. *Environ Res* 153:191–204
- Maghsoudi E et al (2015) Biodegradation of multiple microcystins and cylindrospermopsin in clarifier sludge and a drinking water source: Effects of particulate attached bacteria and phycocyanin. *Ecotoxicol Environ Saf* 120:409–417
- Manubolu M et al (2018) Optimization of extraction methods for quantification of microcystin-LR and microcystin-RR in fish, vegetable, and soil matrices using UPLC-MS/MS. *Harmful Algae* 76:47–57
- Melaram R, Newton AR, Chafin J (2022) Microcystin contamination and toxicity: implications for agriculture and public health. *Toxins* 14(5):350
- Meneely JP, Elliott CT (2013) Microcystins: measuring human exposure and the impact on human health. *Biomarkers* 18(8):639–649
- Miller MJ, Critchley MM, Hutson J, Fallowfield HJ (2001) The adsorption of hepatotoxins from water onto soil during batch studies. *Water Res* 35:1461–1468
- Miller MF, Fallowfield HJ (2001) Degradation of cyanobacterial hepatotoxins in batch experiments. *Water Sci Technol* 43(12):229–232
- Mishra S, Kumar R, Kumar M (2023) Use of treated sewage or wastewater as an irrigation water for agricultural purposes- Environmental, health, and economic impacts. *Total Environ Res Themes* 6:100051
- Mohamed ZA et al (2021) Fungal biodegradation and removal of cyanobacteria and microcystins: potential applications and research needs. *Environ Sci Pollut Res Int* 28(28):37041–37050
- Mohamed ZA et al (2024a) Microcystin levels in irrigation water and field-vegetable plants, and food safety risk assessment: a case study from Egypt. *Toxicon* 247:107846
- Mohamed ZA et al (2024b) Accumulation of microcystin toxin in irrigation water and alfalfa (*Medicago sativa*) forage plant, and assessing the potential risk to animal health. *Chemosphere* 364:143248
- Mohamed ZAAM, Shehri Al (2009) Microcystin-producing blooms of *Anabaenopsis arnoldi* in a potable mountain lake in Saudi Arabia. *FEMS Microbiol Ecol* 69(1):98–105
- Mohammad Rusan MJ, Hinnawi S, Rousan L (2007) Long term effect of wastewater irrigation of forage crops on soil and plant quality parameters. *Desalination* 215(1):143–152
- Musa Yahaya S, Ahmad Mahmud A, Abdu N (2023) The use of wastewater for irrigation: pros and cons for human health in developing countries. *Total Environ Res Themes* 6:100044
- Mutoti M, Gumbo J, Jideani AIO (2022) Occurrence of cyanobacteria in water used for food production: a review. *Physics and chemistry of the earth. Parts a/b/c* 125:103101
- Máthé C et al (2009) Microcystin-LR induces abnormal root development by altering microtubule organization in tissue-cultured common reed (*Phragmites australis*) plantlets. *Aquat Toxicol* 92(3):122–130
- Omidi A, Esterhuizen-Londt M, Pflugmacher S (2018) Still challenging: the ecological function of the cyanobacterial toxin microcystin—what we know so far. *Toxin Rev* 37(2):87–105
- Paerl HW, Paul VJ (2012) Climate change: links to global expansion of harmful cyanobacteria. *Water Res* 46(5):1349–1363
- Pantelic D et al (2013) Cyanotoxins: Characteristics, production and degradation routes in drinking water treatment with reference to the situation in Serbia. *Chemosphere* 91(4):421–441
- Pappas D et al (2020) Beyond microcystins: Cyanobacterial extracts induce cytoskeletal alterations in rice root cells. *Int J Mol Sci* 21(24):1–27
- Petrou M et al (2020) Irrigation of radish (*Raphanus sativus* L.) with microcystin-enriched water holds low risk for plants and their associated rhizospheric and epiphytic microbiome. *Environ Pollut* 266:115208–115208
- Peuthert A, Chakrabarti S, Pflugmacher S (2007) Uptake of microcystins-LR and -LF (cyanobacterial toxins) in

- seedlings of several important agricultural plant species and the correlation with cellular damage (lipid peroxidation). *Environ Toxicol* 22(4):436–442
- Pflugmacher S et al (2019) Translocation of the cyanobacterial toxin microcystin-LR into guttation drops of *Triticum aestivum* and remaining toxicity. *Environ Pollut* (253):61–67
- Pflugmacher S, Aulhorn M, Grimm B (2007) Influence of a cyanobacterial crude extract containing microcystin-LR on the physiology and antioxidative defence systems of different spinach variants. *New Phytol* 175(3):482–489
- Pham T-L, Utsumi M (2018) An overview of the accumulation of microcystins in aquatic ecosystems. *J Environ Manag* 213:520–529
- Pindihama GK, Gitari MW (2020) Cyanobacterial toxins: an emerging threat in South African irrigation water. *Water Environ j: WEJ* 34(3):506–516
- Pinho LX et al (2015) Effect of TiO_2 photocatalysis on the destruction of *Microcystis aeruginosa* cells and degradation of cyanotoxins microcystin-LR and cylindrospermopsin. *Chem Eng J* 268:144–152
- Pouria S et al (1998) Fatal microcystin intoxication in haemodialysis unit in Caruaru. Brazil the Lancet (British Edition) 352(9121):21–26
- Preece EP et al (2017) A review of microcystin detections in estuarine and marine waters: environmental implications and human health risk. *Harmful Algae* 61:31–45
- Prieto A et al (2011) Effects on growth and oxidative stress status of rice plants (*Oryza sativa*) exposed to two extracts of toxin-producing cyanobacteria (*Aphanizomenon ovalisporum* and *Microcystis aeruginosa*). *Ecotoxicol Environ Saf* 74(7):1973–1980
- Radcliffe JC (2022) Current status of recycled water for agricultural irrigation in Australia, potential opportunities and areas of emerging concern. *Sci Total Environ* 807(Pt 2):151676–151676
- Radcliffe JC, Page D (2020) Water reuse and recycling in Australia — history, current situation and future perspectives. *Water Cycle* 1:19–40
- Rastogi RP, Sinha RP, Incharoensakdi A (2014) cyanotoxin-microcystins: current overview. *Rev Environ Sci Biotechnol* 13(2):215–249
- Redouane EM et al (2019) Mode of action and fate of microcystins in the complex soil-plant ecosystems. *Chemosphere* 225:270–281
- Redouane EM et al (2021) Role of rhizospheric microbiota as a bioremediation tool for the protection of soil-plant systems from microcystins phytotoxicity and mitigating toxin-related health risk. *Microorganisms* (Basel) 9(8):1747
- Redouane EM et al (2023) Health risk assessment of lake water contaminated with microcystins for fruit crop irrigation and farm animal drinking. *Environ Sci Pollut Res Int* 30(33):80234–80244
- Romanis CS, Pearson LA, Neilan BA (2021) Cyanobacterial blooms in wastewater treatment facilities: significance and emerging monitoring strategies. *J Microbiol Methods* 180:106123
- Romero-Oliva CS et al (2014) Accumulation of microcystin congeners in different aquatic plants and crops – a case study from lake Amatitlán. *Guatemala Ecotoxicol Environ Saf* 102(1):121–128
- Sangolkar LN, Maske SS, Chakrabarti T (2006) Methods for determining microcystins (peptide hepatotoxins) and microcystin-producing cyanobacteria. *Water Res* 40(19):3485–3496
- Saqrane S et al (2009) Physiological changes in *Triticum durum*, *Zea mays*, *Pisum sativum* and *Lens esculenta* cultivars, caused by irrigation with water contaminated with microcystins: a laboratory experimental approach. *Toxicon* 53(7):786–796
- Sivonen K, Jones G (1999) Cyanobacterial toxins. *Toxic Cyanobact Water: Guide Public Health Consequ, Monitor Manag* 1:43–112
- Spoof L et al (2020) Elimination of cyanobacteria and microcystins in irrigation water—effects of hydrogen peroxide treatment. *Environ Sci Pollut Res Int* 27(8):8638–8652
- Statistics ABo (2014) Australian health survey: nutrition first results - foods and nutrients
- Thomas KF, Boshier H, Cromar C Residential water recycling: Quantity and quality aspects. In: *Water recycling Australia*. Dillon P, (ed). 2000: Commonwealth Science and Industrial Research Organisation & Australian Water Association, Dickson, ACT 2602, Australia p 17-21
- Tsoumalakou E et al (2021) Spray irrigation with microcystins-rich water affects plant performance from the microscopic to the functional level and food safety of spinach (*Spinacia oleracea* L.). *Sci Total Environ* 789:147948–147948
- WHO (2020) Development of WHO guidelines for drinking-water quality and guidelines for safe recreational water environments. World Health Organisation
- WHO (2017) ed. Guidelines for drinking-water quality, 4th ed, incorporating the 1st addendum. World Health Organization. ed. W.H.O Geneva
- Wan X et al (2019) Combined toxic effects of microcystin-LR and phenanthrene on growth and antioxidant system of duckweed (*Lemna gibba* L.). *Ecotoxicol Environ Saf* 185:109668
- Weralupitiya C et al (2022) Cyanotoxins uptake and accumulation in crops: phytotoxicity and implications on human health. *Toxicon* 211:21–35
- Wijewickrama MM, Manage PM (2019) Accumulation of microcystin-LR in grains of two rice varieties (*Oryza sativa* L.) and a leafy vegetable, ipomoea aquatica. *Toxins* 11(8):432
- Wood SA et al (2011) Switching toxin production on and off: intermittent microcystin synthesis in a microcystis bloom. *Environ Microbiol Rep* 3(1):118–124
- Wu X et al (2015) Evaluation of the potential of anoxic biodegradation of intracellular and dissolved microcystins in lake sediments. *J Hazard Mater* 286:395–401
- Xiang A et al (2014) An aptamer-based immunoassay in microchannels of a portable analyzer for detection of microcystin-leucine-arginine. *Talanta* 130:363–369
- Xiang L et al (2020) Bioaccumulation and phytotoxicity and human health risk from microcystin-LR under various treatments: a pot study. *Toxins* 12(8):523
- Xie G et al (2021) Light-driven breakdown of microcystin-LR in water: A critical review. *Chem Eng J* 417:129244

- Yan Q et al (2021) Irrigation with secondary municipal-treated wastewater: Potential effects, accumulation of typical antibiotics and grain quality responses in rice (*Oryza sativa* L.). *J Hazard Mater* 410:124655–124655
- Yerli C et al. (2024) Fertility and heavy metal pollution in silage maize soil irrigated with different levels of recycled wastewater under conventional and no-tillage practices. *Irrigation science* 42:1–18
- Zastepa A, Pick FR, Blais JM (2014) Fate and persistence of particulate and dissolved microcystin-LA from microcystis blooms. *Hum Ecol Risk Assess* 20(6):1670–1686
- Zhang S et al (2021) The latest advances in the reproductive toxicity of microcystin-LR. *Environ Res* 192:110254
- Zhou M, Tu W-W, Xu J (2015) Mechanisms of microcystin-LR-induced cytoskeletal disruption in animal cells. *Toxicol* 101:92–100
- Zhu J et al (2018) Effect of irrigation with microcystins-contaminated water on growth and fruit quality of *Cucumis sativus* L. and the health risk. *Agric Water Manag* 204:91–99
- Žegura B et al (2011) Microcystin-LR induced DNA damage in human peripheral blood lymphocytes. *Mutat Res Genet Toxicol Environ Mutagen* 726(2):116–122

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